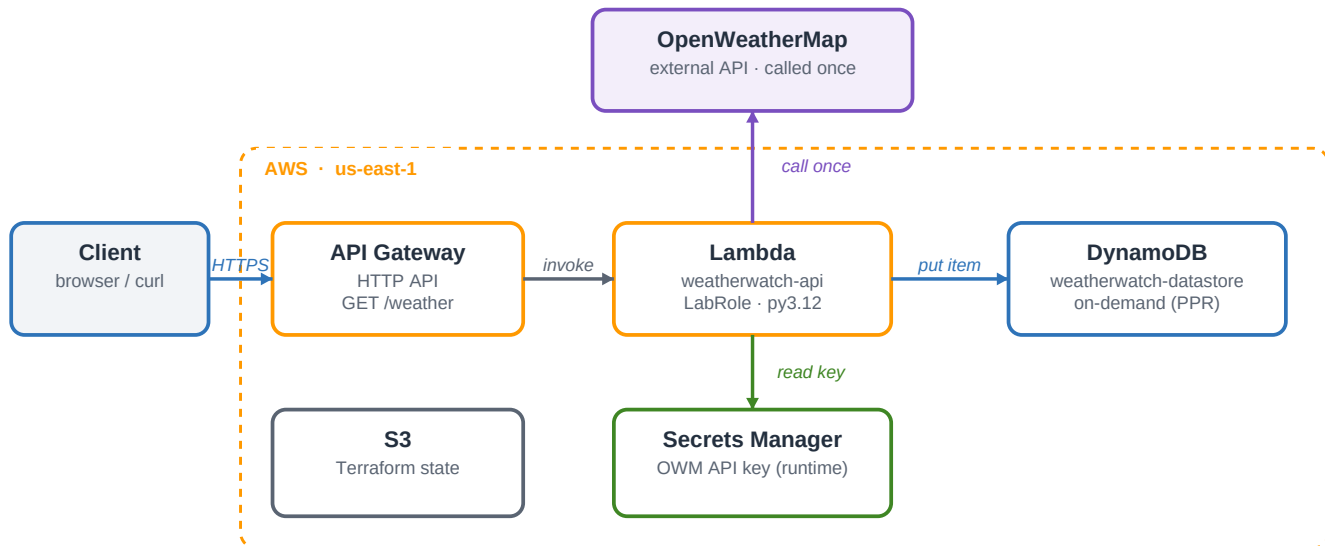


Architecture & Key Decisions

A serverless, scale-to-zero weather poller: a public HTTPS endpoint invokes a Lambda that reads its API key from Secrets Manager, calls OpenWeatherMap once, transforms the result, and stores it in an on-demand DynamoDB table.



Request path: **Client** → **API Gateway (HTTP API)** → **Lambda (LabRole)**. Inside the handler: read key from **Secrets Manager** → call **OpenWeatherMap** once → **PutItem** into **DynamoDB**. Terraform state lives in **S3**. All in us-east-1.

Three decisions

On-demand DynamoDB over RDS (PostgreSQL) RDS wins on ad-hoc SQL and joins, but bills for a running instance ~24×7 even while idle. PAY_PER_REQUEST DynamoDB costs \$0 at rest — right for a bursty, mostly-idle poller with simple key access.	COST
API Gateway HTTP API over Lambda Function URL (AWS_IAM) A Function URL is one less resource, but every request must be SigV4-signed, so it can't be opened or tested in a browser. HTTP API gives a clean, public, browser-testable HTTPS front door. (A NONE-auth Function URL is blocked by the lab SCP.)	OPERATIONAL EXCELLENCE
AWS Secrets Manager over Lambda env var / hard-coded key An env var is free and simpler, but the key then sits in plaintext in the function config and in Terraform state. Secrets Manager keeps it out of code and state, read only at runtime, and supports rotation (~\$0.40/secret/mo).	SECURITY